With data, fibre entry is equivalent to

$$|\beta| \leqslant 33.75°$$

Consider the three rays in turn.

(i) $\alpha_1 = 30°$:

a. Can the refracted ray enter the fibre?

Using (2), $\sin\theta_1 = 2\sin\alpha_1 = 1$; hence

$$\theta_1 = 90°$$

Substitute into (3):

$$\theta_1' = \arcsin\left(\frac{2}{3}\right) = 41.81°$$

Substitute into (4):

$$\beta_1 = 2\theta_1' - \theta_1 + \alpha_1 = 23.62° \qquad (7)$$

Since

$$\beta_1 < \angle QOB \qquad (8)$$

the ray enters the fibre.

b. Can total internal reflection occur at the core - cladding interface? Let the critical angle be i_C, then

$$i_C = \arcsin\left(\frac{1.400}{1.500}\right) = 68.96° \qquad (9)$$

By $i = 90° - \alpha'$, total reflection requires α' to satisfy the following:

$$\alpha' \leqslant 90° - 68.96° = 21.04° \qquad (10)$$

Now $\alpha_1' = \theta_1 - \alpha_1 - \theta_1' = 18.19°$ meets (10); thus it propagates by total internal reflection in the fibre.

(ii) $\alpha_2 = 25°$:

Using (2): $\sin\theta_2 = 2\sin\alpha_2 = 2\sin25°$ $\theta_2 = 57.70°$

$$\sin\theta_2' = \frac{1}{1.5}\sin\theta_2 \qquad \theta_2' = 34.30°$$

$$\beta_2 = 2\theta_2' - \theta_2 + \alpha_2 = 35.90° \qquad (11)$$

(6): ray cannot enter the fibre right of QQ'.

(iii) $\alpha_3 = 20°$:

Using (2): $\sin\theta_3 = 2\sin\alpha_3 = 2\sin20°$ $\theta_3 = 43.16°$

$$\sin\theta_3' = \frac{1}{1.5}\sin\theta_3 \qquad \theta_3' = 27.13°$$

$$\beta_3 = 2\theta_3' - \theta_3 + \alpha_3 = 31.10° \qquad (12)$$

(6): ray enters the fibre right of QQ'.

Then

$$\alpha_3' = \theta_3 - \alpha_3 + \theta_3' = -3.97° \qquad (13)$$

Negative: backward ray meets the axis left of A, hitting the upper boundary from below (Fig. 4). $|\alpha_3'| < 21.04°$; it can propagate in the fibre by total internal reflection.

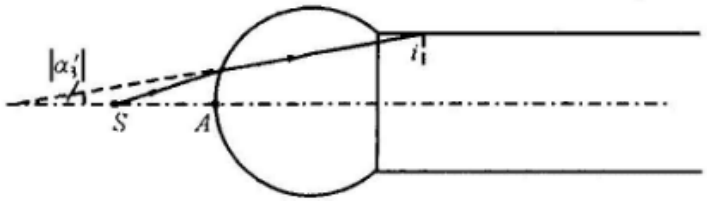


Fig. 4